

WILDFIRE RISK CLASSIFICATION

Using ML Models - SVM, Random Forest & XGBoost

Presented By:

Anagha Jayarajan (253201)

M R Lekshmi Priya (253011)

Akshay B S (253014)

PROBLEM STATEMENT

- Predict wildfire risk level for a given location and environmental conditions.
- Target classes:
 - Low (●): Minimal fire activity; low FRP and benign meteorological conditions.
 - Medium (●): Moderate conditions with potential for fire ignition/spread.
 - High (●): Elevated fire danger — high FRP, high temperature, low humidity, strong winds.
- Success criteria:
 - High F1-macro and balanced precision/recall across classes (minimize missed high-risk areas and false high-risk alarms).
 - Robust performance on spatially separated test sets (generalizes to new regions).

DATASET — NASA FIRMS

- Source: NASA FIRMS (Fire Information for Resource Management System) — public active-fire satellite dataset (MODIS/VIIRS).
- Data Types: Satellite-derived detections + ancillary meteorological variables.
- Key Features used in this project:
 - Spatial: Latitude, Longitude
 - Satellite-derived: Brightness, Brightness_T31, Scan, Track, FRP (Fire Radiative Power)
 - Detection: Confidence, Day/Night indicator, Detection time (timestamp)
 - Meteorological / Environmental: Temperature, Humidity, Wind Speed, Drought Index (where available)
 - Metadata: Source sensor, acquisition time
- Usage in project:
 - Basis for feature matrix and target labels for SVM, Random Forest, and XGBoost
 - Spatial split used to evaluate geographic generalization
- Key EDA takeaway: FRP and meteorological variables are primary predictors; spatial clustering and class imbalance require careful splitting and weighting.

PREPROCESSING & FEATURE ENGINEERING

- Data cleaning:
 - Standardize column names and parse timestamps
 - Remove duplicates, invalid coordinates, and extreme outliers
 - Convert numeric-like strings to numeric types
- Missing value handling:
 - Use SimpleImputer (median for numeric, most_frequent for categorical)
 - Drop rows with irrecoverable missing coordinates or timestamps
- Encoding & scaling:
 - One-hot encode categorical flags (day/night, sensor)
 - Label-encode target classes
 - Scale numeric inputs (StandardScaler) inside a ColumnTransformer pipeline

- Feature creation:
 - Derive risk bins from FRP / composite index (Low/Medium/High)
 - Temporal features: hour, day-of-year, season
 - Spatial features: lat, lon; optional clustering (region id)
 - Interaction features: FRP × temperature, humidity × wind_speed
- Pipeline & persistence:
 - Build preprocessing pipeline with ColumnTransformer + Pipeline
 - Fit on training data; save fitted scaler, encoder, and pipeline with joblib
 - Ensure identical preprocessing during deployment (Streamlit app)
- Spatially-aware data splitting: Create geographically disjoint train/test sets to prevent leakage (e.g., cluster coordinates and hold out clusters, or use GroupShuffleSplit with region IDs)
- Rationale: Neighboring coordinates are correlated; random splits overestimate generalization
- Stratification: Stratify splits by target class to preserve class proportions where possible

SUPPORT VECTOR MACHINE (SVM)

- Type: Kernel-based classifier (e.g., RBF kernel)
- Use: Baseline high-capacity classifier for tabular features
- Pros: Good decision boundary flexibility; strong when features are well-separated
- Cons: Sensitive to scale, slow on large datasets, and prone to leakage/overfitting with spatial data
- Typical settings: C and gamma grid search; standardize inputs before training
- Notes from project: Achieved very high accuracy (~99.5%) but likely affected by target leakage or overly easy random split

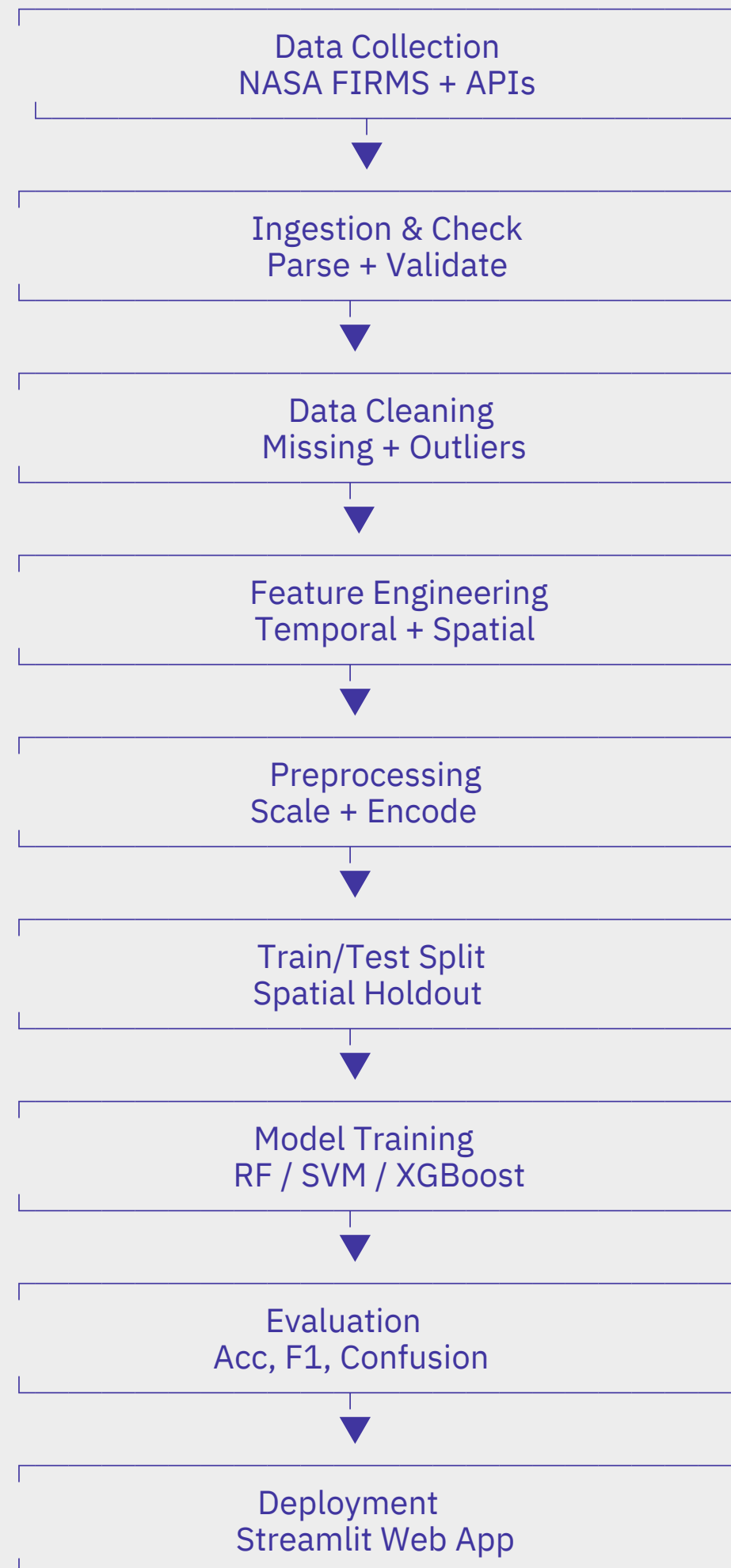
RANDOM FOREST CLASSIFIER

- Type: Ensemble of decision trees (bagging)
- Use: Interpretable baseline with built-in feature importance
- Pros: Handles non-linearities, robust to noisy features, minimal scaling required
- Cons: Can underperform gradient-boosted methods on complex patterns; may need many trees for stability
- Typical settings: number of trees (n_estimators), max depth, class weighting for imbalance
- Notes from project: Good interpretability; lower performance (~77.4% accuracy) relative to XGBoost

XGBOOST CLASSIFIER

- Type: Gradient-boosted decision trees
- Use: Final production model for best accuracy and spatial generalization
- Pros: Excellent predictive performance, handles missing values, captures complex feature interactions, fast with optimized implementation
- Cons: More hyperparameters to tune; risk of overfitting without proper CV
- Typical settings: `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, early stopping on validation
- Notes from project: Best overall — 92.2% accuracy and F1-macro 0.922; robust under spatial train/test splits, selected for deployment

PROJECT PIPELINE



PREVIEW

Deployed via Streamlit
Link: <https://wildlife-risk-classification-hmhgxew5ym8wycpsdoegi8.streamlit.app/>

📌 Input Parameters

🌐 Location

Latitude

10.00 - +

Longitude

76.00 - +

🛰️ Satellite Info

Satellite ?

T ▾

Day / Night ?

D ▾

📡 Scan Parameters

Scan

1.50

Track

1.50

🔥 Thermal & Fire Parameters

Share ☆ ⋮

🔥 Wildfire Risk Classification System

Satellite & Weather Based Wildfire Prediction using XGBoost

Predicts wildfire risk levels — ● Low · ● Medium · ● High — using NASA FIRMS MODIS satellite observations.

> 📄 View Input Data

Predict Wildfire Risk

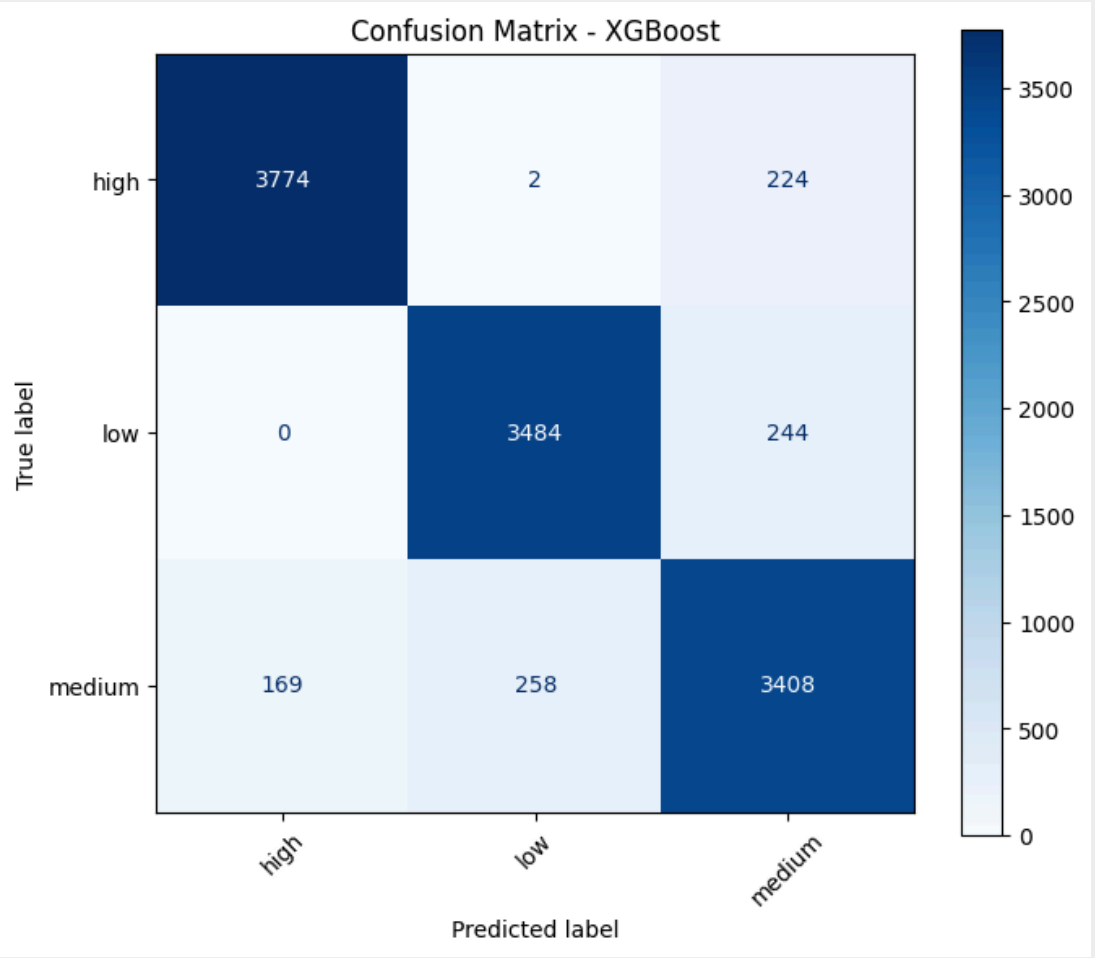
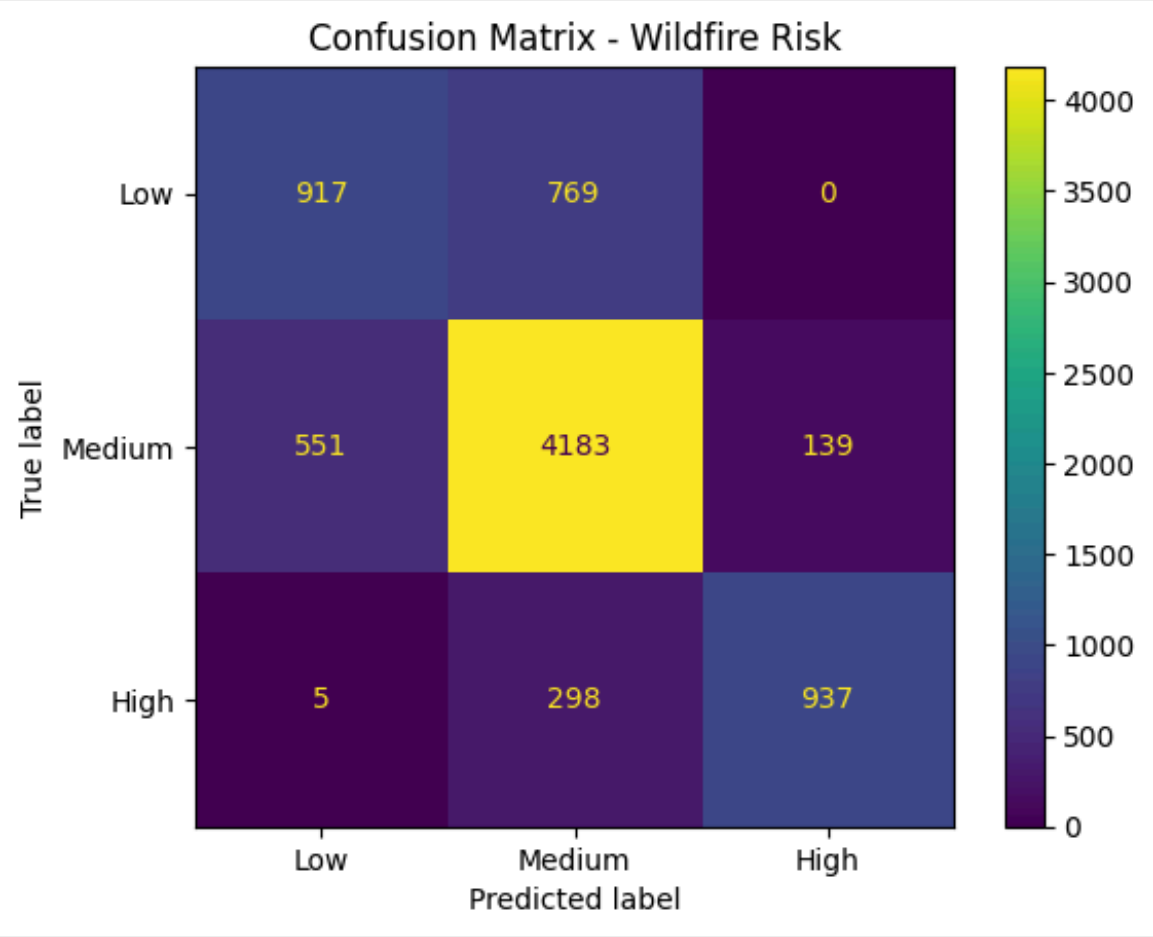
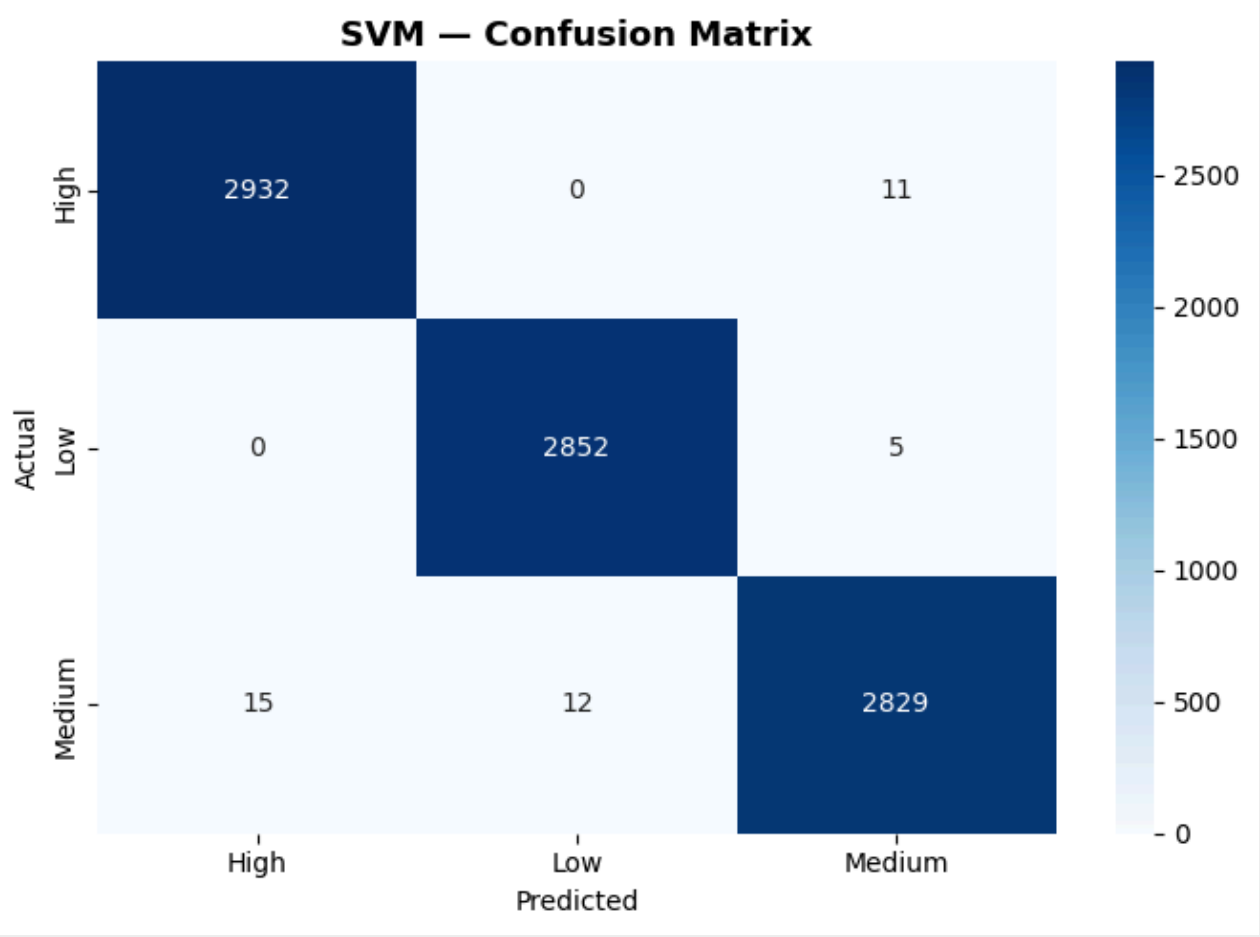
> ⓘ About This Project

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MODEL EVALUATION

- SVM — Accuracy: 99.5% | F1-Macro: 0.995 — likely target leakage / overly easy random split
 - Random Forest — Accuracy: 77.4% | F1-Macro: 0.74 — Interpretable baseline, lower predictive power
 - XGBoost — Accuracy: 92.2% | F1-Macro: 0.922 — Best balance of accuracy and robustness
- Key takeaway: XGBoost maintains high, balanced performance while Random Forest underfits and SVM's near-perfect score is suspicious.

CONFUSION MATRICES



CONCLUSION

- Project outcome: Built a reproducible pipeline to predict wildfire risk (Low / Medium / High) using NASA FIRMS satellite and meteorological data.
 - Best model: XGBoost chosen for production — 92.2% accuracy and 0.922 F1-Macro, with good spatial generalization.
 - Reliability note: SVM's near-perfect accuracy likely indicates data leakage; spatial train/test splits produce more realistic evaluation.
 - Limitations: Data limited to FIRMS, class imbalance, and static models (no streaming/temporal retraining).
- Next steps:
 - Integrate live weather feeds and real-time satellite ingestion.
 - Add temporal models (LSTM/CNN) for spread prediction.
 - Improve spatial visualization (Folium/Kepler.gl) and mobile delivery.
 - Establish continuous retraining + monitoring for model drift.

The image features a light gray background with decorative purple geometric shapes in the corners. In the top-left corner, there are two overlapping triangles, one in a darker shade of purple and one in a lighter shade. In the bottom-right corner, there are also two overlapping triangles, one in a darker shade of purple and one in a lighter shade.

**THANK
YOU**